Exam Date & Time: 25-Sep-2023 (08:00 AM - 10:00 AM)

MANIPAL ACADEMY OF HIGHER EDUCATION

BTech III Semester Sessional Examination 2023
COMPUTER ORGANIZATION AND ARCHITECTURE [CSE 2121]

Duration: 120 mins

Multiple Choice Questions:

Q1. Which one of the following integers could be represented in 2s complement but not in 1s complement and sign – magnitude systems, the number of bits being 5? (0.5)

1. +16
2. **-16**
3. +15
4. -15

Q2.The flags/condition codes that are affected as a result of execution of the following instruction are

Add R3,#1

Assume that R3 is an 8-bit register, and it has the decimal value 255 in it.

(0.5)

1. O and Z.
2. **O, C and Z.**
3. Only Z.
4. None of them are affected.

Q3. Which one of the following is performed in Rounding technique when the guard bits G=1, R=0, and S=0? (0.5)

1. Round up
2. Round down
3. Round to nearest odd
4. **Round to nearest even**

Q4. When 1.110001 is truncated to 3 binary places as per Von Neumann rounding, there is a _____ with a value of _____. (0.5)

1. loss, 0.875
2. **gain, 0.875**
3. loss, 0.125
4. gain, 0.125

Q5. How many add operations are performed in multiplication process to reach final product if the multiplicand B is 10010 and the multiplier Q is 10110? (0.5)

1. 4
2. 2
3. 5
4. **3**

Q6. In the Booth's multiplication algorithm, after initialization the ---------- bit of the multiplier Q_n (where n is the no of bits in the multiplier) is tested. (0.5)

1. higher order
2. sign bit
3. **lower order**
4. Both 1 and 2

Q7. Which addressing mode executes its instructions within the CPU without the necessity of referencing the memory for operands? (0.5)

1. Implied Mode
2. Immediate Mode
3. Base with Index Mode
4. **Register Mode**

Q8. In the following indexed addressing mode instruction,

Move 10(R1), LOC

the effective address is _____ (0.5)

1. 10 + R1
2. 10 * R1
3. [10 + R1]
4. **10 + [R1]**

Q9. What is the name of the addressing mode where EA = [Ri] + [Rj]? (0.5)

1. **Base with Index**
2. Register Indirect
3. Register Absolute
4. Indexed

Q10. Which of the following is not a CISC instruction set? (0.5)

1. Move C, B

   Add C, A

2. Move Ri, A

   Add Ri, B

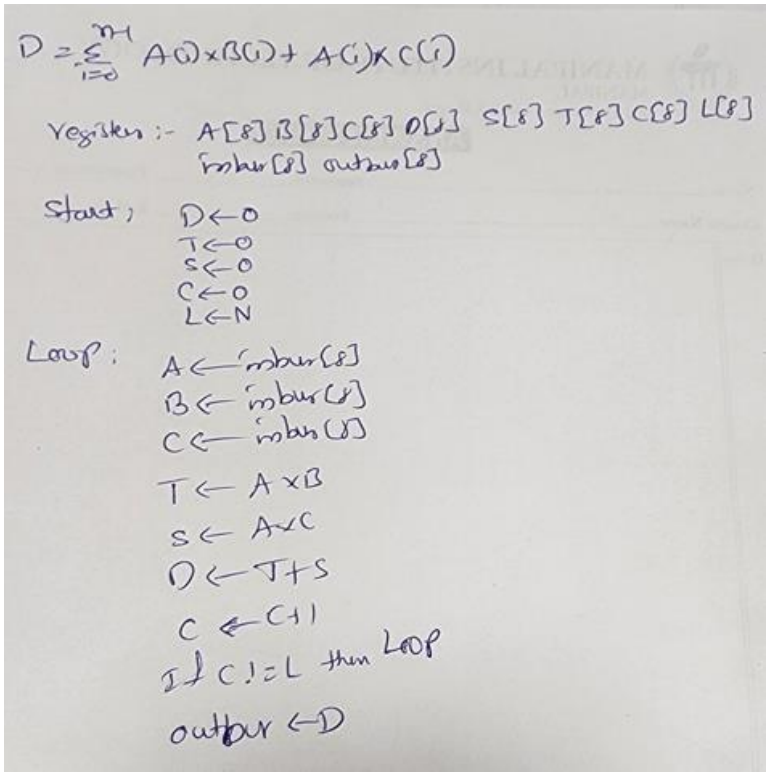
3. Move ITEM, (SP)+

4. **Load R6, R2**

Descriptive Type Questions:

Q11. For the following task, give the complete design of processing section establishing control points. Assume that all inputs and outputs are of same bit size and any loop condition or comparator output if needed can be assumed to be one bit of status register. Assume a single iteration of control signals for the solution. (4)

$$D = \sum_{i=0}^{n-1} A[i] * B[i] + A[i] * C[i]$$

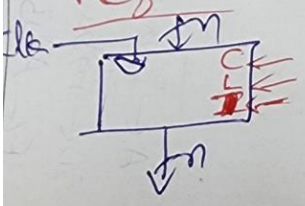


C ← C+1

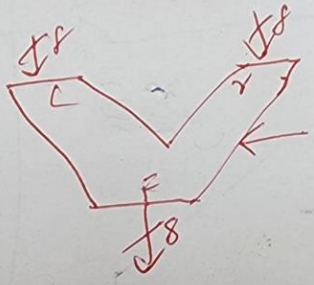
If C != L then LOOP

outbus ← D

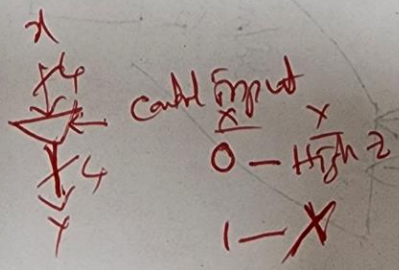
Register

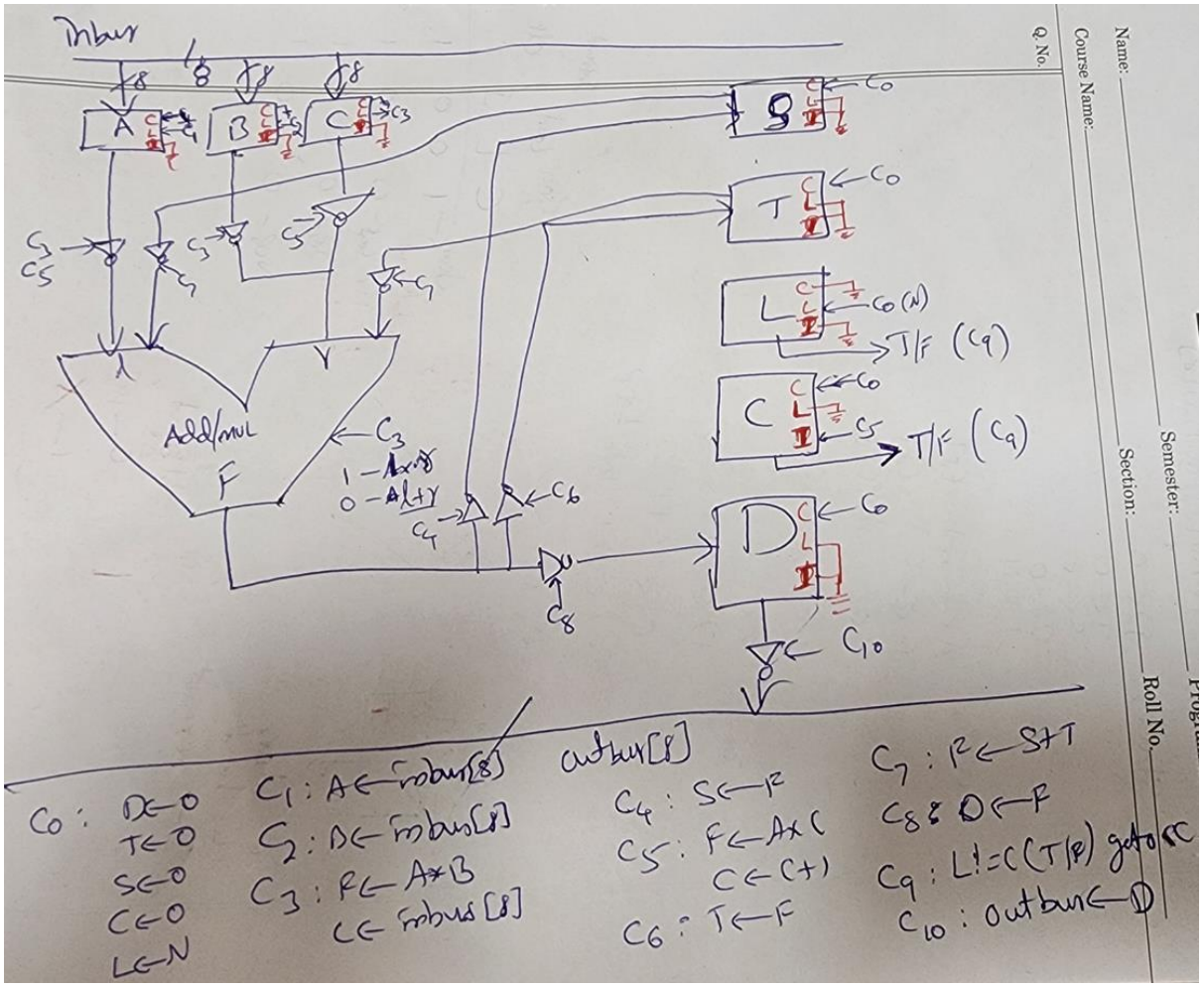


C = clear
L = load
I = Increment



	Input		
	clear	load	clk
	C	L	I
clear ←	1	0	0
Load –	0	1	0
Inc –	0	0	1
NOP –	0	0	0





C0: D←0, T←0, S←0, C←0, L←N

C1: A←Inbus[8]

C2: D←Inbus[8]

C3: R←A×B, C←Inbus[8]

C4: S←R

C5: F←A×C, C←(C+1)

C6: T←F

C7: R←S+T

C8: D←R

C9: L!=C (T/F) goto C

C10: Outbus←D

outbus[8]

Q. No.

Name: _____ Semester: _____

Course Name: _____ Section: _____

Roll No. _____

Q12. Write a CISC style program to find the sum of even numbers in a list. Assume that there is an instruction called MOD that yields the remainder of a division operation. Make a table to map all the operands in your program to appropriate addressing modes. (4)

12)
```
    Move  R0, N
    clear R1
    Move  R2, #LIST
LOOP: Move  R3, (R2)+   ---> Move R4, R3
    Mod   R4, #2
    Branch != 0  L1
    Add   R1, R3
L1: Subtract R0, #1
    Branch > 0  LOOP
    Move  SUM, R1
```

// 2.5 M

1)	R0 ⇒ Register direct N ⇒ Absolute	6	~~L1 may be Absolute~~
2)	R1 ⇒ Register direct	7	R1, R3 } Register direct
3)	R2 ⇒ Register direct #LIST ⇒ Immediate	8	R0 ⇒ Register direct #1 ⇒ Immediate
4)	R3 ⇒ Register direct (R2)+ ⇒ Autoincrement	9	
5	R4 ⇒ Register direct #2 ⇒ Immediate	10	SUM ⇒ Absolute R1 ⇒ Register direct

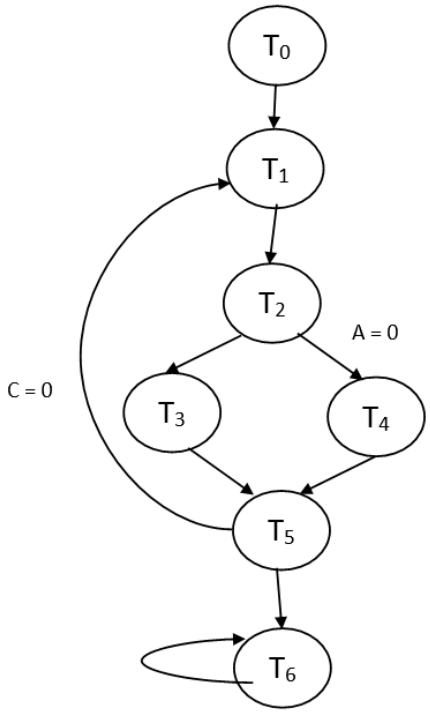
// 1.5 M

Q13. For the following State Transition Diagram

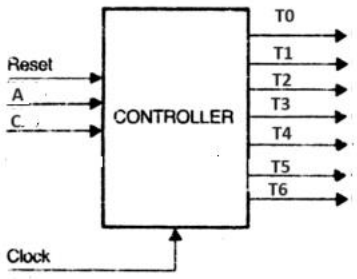
A) Draw the block diagram of the controller

B) Draw the logic diagram of the controller indicating appropriate counter, decoder and sequence controller

C) Derive the truth table of the sequence controller and provide expressions for the output terms of the sequence controller (3)

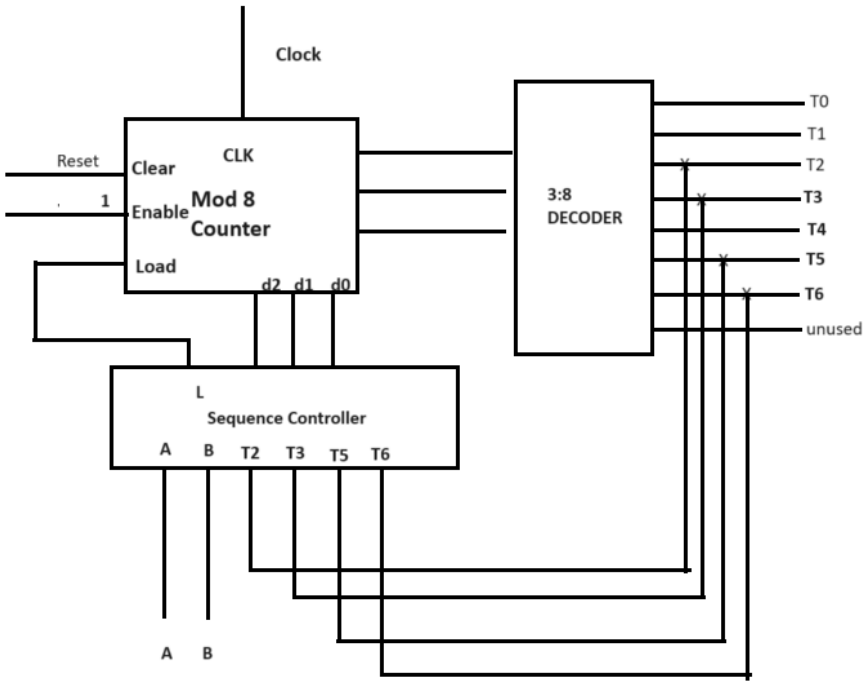


SOLUTION:



A) **0.5 M**

B)



1M

C)

CLK	A	C	T2	T3	T5	T6	L	d2	d1	d0
↓	0	X	1	X	X	X	1	1	0	0
↓	X	X	X	1	X	X	1	1	0	1
↓	X	0	X	X	1	X	1	0	0	1
↓	X	X	X	X	X	1	1	1	1	0

1.5M

Q14. Perfrom Twos Complement Multiplication of +45 and +63 using Booth's Algorithm showing all the steps in detail. Indicate the product and provide its integer value. (3)

+45 = 0101101 **(M)** **-M =** 1010011

+63 = 0111111 **(Q)** **0.5M**

A	Q	Q_{-1}	M	
0000000	0111111	**0**	0101101	
1010011 1101001	0111111 1011111	0 **1**	0101101	A=A-M ASR(A,Q)
1110100	1101111	**1**	0101101	ASR(A,Q)

1111010	0110111	**1**	0101101	ASR(A,Q)
1111101	0011011	**1**	0101101	ASR(A,Q)
1111110	1001101	**1**	0101101	ASR(A,Q)
1111111	0100110	**1**	0101101	ASR(A,Q)
1111111 + 0101101 = 0101100 **0010110**	0100110 **0010011**	1	0101101	A=A+M ASR(A,Q)

2M

Final Product is (A,Q) = **00101100010011 =** $(2835)_{10}$ **0.5M**

Q15. Using Restoring Division Algorithm, divide 342 by 57 showing all the steps in detail. Indicate the quotient and remainder of this division and provide the integer values. (3)

342 = 0101010110 **(Q)**

57 = 0000111001 **(M)** **-M =** 1111000111 **0.5M**

A	**Q**	
0000000000	0101010110	
0000000000 1111000111 0000000000	101010110_ 1010101100	LS(A,Q) A=A-M <0 Restore A, Q_0 = 0
0000000001 1111001000 0000000001	010101100_ 0101011000	LS(A,Q) A=A-M < 0 Restore A, Q_0 = 0
0000000010 1111001001 0000000010	101011000_ 1010110000	LS(A,Q) A=A-M < 0 Restore A, Q_0 = 0
0000000101 1111001100 0000000101	010110000_ 0101100000	LS(A,Q) A=A-M < 0 Restore A, Q_0 = 0
0000001010 1111010001	101100000_	LS(A,Q) A=A-M < 0

0000001010	1011000000	Restore A, $Q_0 = 0$
0000010101 1111011100 0000010101	011000000_ 0110000000	LS(A,Q) A=A-M < 0 Restore A, $Q_0 = 0$
0000101010 1111110001 0000101010	110000000_ 1100000000	LS(A,Q) A=A-M < 0 Restore A, $Q_0 = 0$
0001010101 0000011100 	100000000_ 1000000001	LS(A,Q) A=A-M > 0 $Q_0 = 1$
0000111001 0000000000 	000000001_ 0000000011	LS(A,Q) A=A-M > 0 $Q_0 = 1$
0000000000 1111000111 **0000000000**	000000011_ **000000110**	LS(A,Q) A=A-M < 0 $Q_0 = 0$

2M

Remainder(A) = 0000000000 = 0_{10}

Quotient (Q) = 000000110 = 6_{10} **0.5M**

Q16. Draw the State Diagram for the Register Transfer Description provided below. Provide a table that lists the operations performed and the control signals that have to be activated for each state.

Declare Registers A[8],B[8],C[8],D[8],T[1];

Declare Buses: Inbus1[7], Inbus2[7], Outbus[7];

Start: A[6:0]←Inbus1, B[6:0] ←Inbus2, A[7] ←0,B[7] ←0, D←0,T←0;

C[6:0]←Inbus1, C[7]←0;

D←A-B;

T[0]←D[7];

If T[0]=0 then D←A;

If T[0]=1 then D←B;

D←D-C;

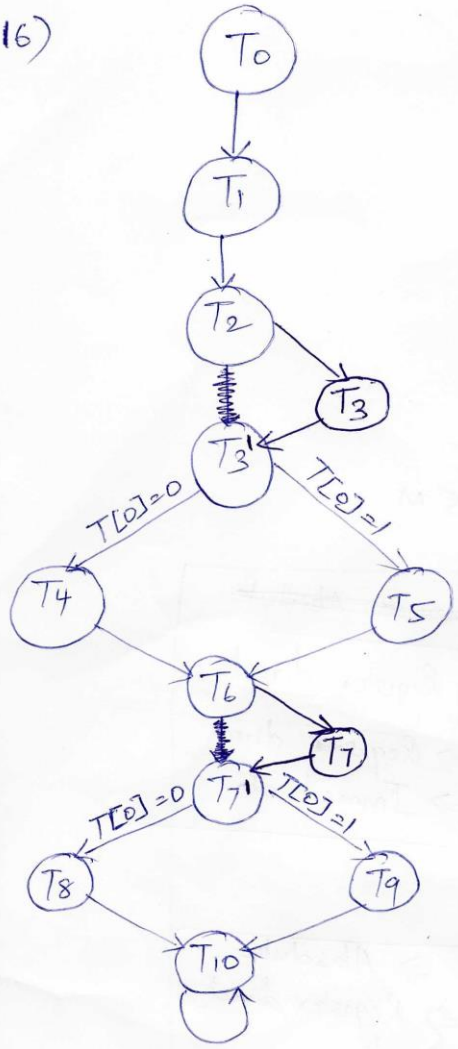
T[0]←D[7];

If T[0] =1 then Outbus=C[6:0];

If T[0]=0 then Outbus=D[6:0];

Halt: Goto Halt (3)

16)



// 2M

		Control points
T_0	$A[6:0] \leftarrow$ inbus 1, $B[6:0] \leftarrow$ inbus 2, $D \leftarrow 0$, $T \leftarrow 0$	C_0, C_1, C_2, C_3^*
T_1	$C[6:0] \leftarrow$ inbus 1	C_4
T_2	$D \leftarrow A - B$	C_5, C_6 $\{C_7, C_8\} = 00$
T_3	$T[0] \leftarrow D[7]$	C_9
T_3'	None	None
T_4	$D \leftarrow A$	C_6 $\{C_7, C_8\} = 01$
T_5	$D \leftarrow B$	C_6 $\{C_7, C_8\} = 10$
T_6	$D \leftarrow D - C$	C_5, C_6 $\{C_7, C_8\} = 00$
T_7	$T[0] \leftarrow D[7]$	C_9
T_7'	None	None
T_8	Outbus = $c[6:0]$	C_{10}
T_9	Outbus = $D[6:0]$	C_{11}
T_{10}	None	None

// 1M.

Assumptions:

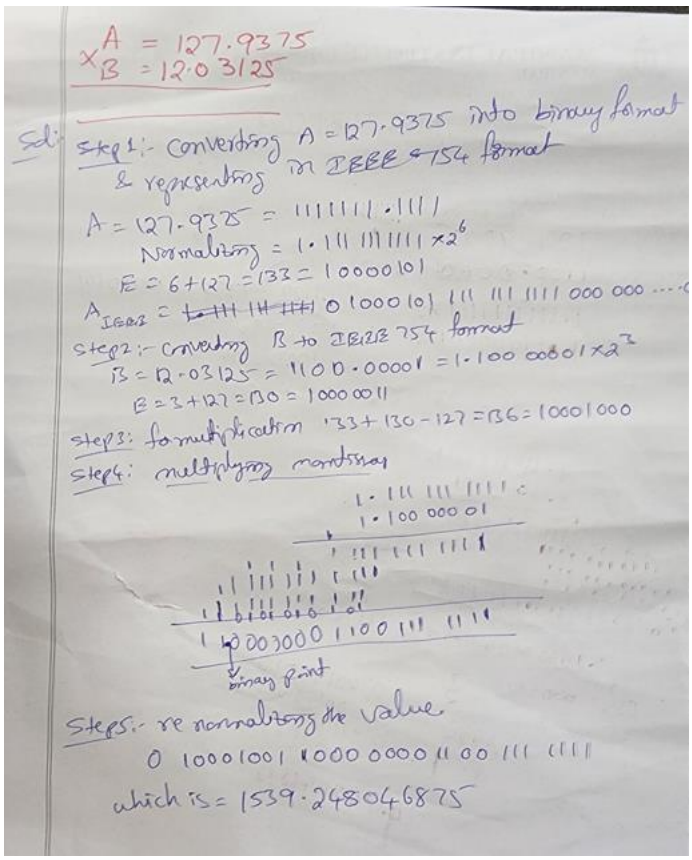
1) C_7, C_8 are selection inputs of 4x1 MUX System in register D.

2) Subtractor has the ability to choose b/w the operands A & B (or)

Q17. Multiply the following two number using IEEE-754 (single precision) floating point representationwith steps in detail.

A=127.9375

B=12.03125 (3)



Q18. List and describe the different types of computers. (2)

0.5 M for each type listing and explanation

1. Embedded computers
2. Personal computers
3. Servers and Enterprise systems
4. Supercomputers and Grid computers